The Amazing Life of Honeybees

Honeybees are small flying insects best known for making honey and for pollinating flowers. A single honeybee colony can contain between 20,000 and 60,000 bees living together in one hive.

Honeybees have been working alongside humans for thousands of years. Ancient Egyptians kept bees in clay hives, and honey was prized as both food and medicine.

This short guide explains how a hive is organised, how bees make honey, how they communicate, and why they matter so much to the food we eat.

Inside the Hive

Every hive has three types of bees. The queen is the only bee that lays eggs, and she can lay up to 2,000 eggs in a single day. A queen bee can live for three to five years.

Worker bees are all female. They clean the hive, feed the young, guard the entrance, and collect nectar and pollen. In summer a worker bee lives for only about six weeks.

Drones are the male bees. Their main job is to mate with a queen from another hive. Drones do not have a stinger and do not collect food. Before winter, the workers push the drones out of the hive.

How Honey Is Made

Worker bees visit flowers and drink a sugary liquid called nectar, which they store in a special honey stomach. Back at the hive, they pass the nectar mouth to mouth to other bees.

The bees then spread the nectar into wax honeycomb cells and fan it with their wings. This removes water until the nectar thickens into honey. When it is ready, the bees seal the cell with a cap of wax.

To make just one kilogram of honey, bees must visit about four million flowers. An average worker bee makes only one twelfth of a teaspoon of honey in her whole life.

The Waggle Dance and Why Bees Matter

When a worker finds a good patch of flowers, she returns to the hive and performs a figure-eight movement called the waggle dance. The angle of the dance shows the direction of the flowers relative to the sun, and the length of the waggle shows how far away they are.

Bees are also essential pollinators. About one third of the food people eat, including apples, almonds, blueberries and cucumbers, depends on pollination by bees and other insects.

Bee populations have been declining because of habitat loss, pesticides and disease. You can help by planting flowers, avoiding pesticides in your garden, and buying honey from local beekeepers.